Fall 2025 Midterm 2

The exam is closed book and closed notes. You are allotted one 8.5" x 11", single sided, "cheat sheet" of handwritten notes. You may also use a simple scientific (non-graphing) calculator. Your phone is not allowed as a calculator. Using any resources outside of the aforementioned items is strictly prohibited.

While you take the exam, you are prohibited from discussing the test with anyone. If you are taking the test after your classmates, you are also prohibited from talking to them about the test before you take it. Evidence of cheating may result in a 0 on the exam and be reported to the Student Conduct Board.

First and Last Name (print clearly): _____

Student ID Number (print clearly): _____

Please read the following statement and sign below to indicate that you understand the policies for this exam

You must show your student ID when you submit your test.

UC Berkeley Honor Code
"As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others." Please carefully read the statements below, and indicate your understanding and intent to adhere to the UC Berkeley Honor code by signing your name in the space below. I agree not to engage in any of the following behaviors:

- Copying or attempting to copy from others during an exam or on an assignment.
- Communicating answers with another person during an exam.
- Pre-programming a calculator or other personal electronic device to contain answers, or using other unauthorized information for exams.
- Using unauthorized materials, i.e. prepared answers.
- Allowing others, or utilizing software like generative AI, to do an assignment or a portion of an assignment for you, including the use of a commercial term-paper service.
- Submitting the same assignment for more than one course without prior approval of all the instructors involved.
- Collaborating on an exam or assignment with any other person without prior approval from an instructor.
- Taking an exam for another person or having someone take an exam for you.
- Altering a previously graded exam or assignment for the purpose of a grade appeal or of gaining points in a re-grading process.
- Submitting an electronic file the student knows to be unreadable or corrupted instead of a completed assignment.

Signature:

Sign your name: _____

INSTRUCTIONS:

Hand write your responses using a pencil or pen in the space provided. Give your responses ONLY in the space provided. Do not write your responses outside of the provided spaces. The additional space provided, including space on the back side of the exam can be used as scratch paper but will not be graded.

Phones should be turned OFF prior to the start of the exam and secured in your backpack or another secure location. Do not leave your phone or other electronic devices out. If you need to leave the room for any reason during the exam please flag a GSI and let them know prior to exiting the room. Upon leaving the exam room, you must leave your electronic device(s) with the GSIs. Time will still accrue when you leave the room.

The length of the midterm is 50 minutes. If you finish early and are satisfied with your work you may leave early. Hand in your exam to a GSI, who will verify that they received it, with your student ID.

- Unless otherwise specified in the question, format your answers according to the following guidelines:
  - present your answers rounded to two decimal places
  - present proportions as decimal values rounded to 2 decimal places rather than percents (ex: 0.41 rather than 41.30%)
- All logarithms ("logs") are the natural logarithm (base e; $log_e(x)$; $ln(x)$)

Exam Format:

Short Format Questions: Q1-5 [5 points]
Long Format Question: Q6 [8 points]
Long Format Question: Q7 [6 points]
Long Format Question: Q8 [6 points]
Long Format Question: Q9 [5 points]
Extra credit: Q10 [1 point]
Optional Feedback Question

Short Format Questions: Please select only one answer choice.

1. [1 point] True or False:

Suppose $P(A) = 0.72$ and $P(B) = 0.55$, and $P(A \cap B) = 0.38$. Therefore, A and B are independent.

☐ A. True
☐ B. False

2. [1 point] Flu cases are on the rise at UC Berkeley, and campus health researchers are recording the number of student visits to the Tang Center for flu-like symptoms each day. They want to model the probability of observing a certain number of cases on *any given day*. Choose the most appropriate theoretical distribution.

☐ A. Normal
☐ B. Binomial
☐ C. Poisson
☐ D. None of the above

3. [1 point] Which statement about the Normal Distribution is correct? Select one answer.

☐ A. The Normal distribution is used to model discrete random variables
☐ B. Under the Standard Normal curve, $P(x = 0) > 0$
☐ C. The Normal distribution is defined by one parameter, μ
☐ D. The sampling distribution of the sample mean is approximately Normal when n is large

4. [1 point] An investigator in the Environmental Health Sciences department collects 12 independent water samples from Strawberry Creek, each with a 0.25 probability of being contaminated with nitrates, a type of chemical contaminant.

Let X represent the number of contaminated samples. They would like to compute $P(X \leq 3)$, where $X \sim Binomial(12, 0.25)$. Which R function should they use?

☐ A. `pbinom()`
☐ B. `dbinom()`
☐ C. `rbinom()`
☐ D. None of the above

5. [1 point] Epidemiologists are investigating whether exposure to Febreze air fresheners increases asthma risk. A Type II error in this study could lead to which of the following? Select one answer.

☐ A. Correctly concluding that Febreze exposure increases asthma risk when it truly does
☐ B. Correctly concluding that Febreze exposure does not increase asthma risk when it truly does not
☐ C. Failing to detect that Febreze exposure truly increases asthma risk
☐ D. Concluding that Febreze exposure increases asthma risk when it actually does not

Long Format Questions

Q6 [8 Points]

The University of California reported a record number of peer-reviewed publications per department this year. Historically, the UC departments publish an average of 96.5 papers per year with a population standard deviation of 8.5 papers.

A random sample of 25 UC departments reported an average of 100.5 papers this year. Researchers want to know whether this provides strong statistical evidence that UC's average productivity has increased.

Assume the number of publications per department is approximately Normally distributed. Use a significance level of $\alpha = 0.05$.

6.1 [2 points] State the null and alternative hypotheses:

H_0 :

H_A :

6.2 [2 points] Calculate the z test statistic.

6.3 [2 points] Write a line of R code that would calculate the p-value. Only use function(s) that assume a standard Normal distribution (i.e., N(0, 1))

6.4 [2 points] The R output below shows the p-value for this test. Interpret this p-value in the context of the problem. What conclusion would you draw about UC's research productivity? (1-2 sentences)

```
## [1] 0.009386706
```

Q7 [6 Points]

The City of Berkeley recently completed the Southside Mobility Improvements Project, adding protected bike lanes and upgraded intersections around campus. City planners estimate that 65% of students now feel safe biking through Southside, 75% feel safe walking, and 55% feel safe doing both.

You decide to randomly survey 8 students who travel through Southside each day.

7.1 [1 point] What is the probability that a randomly selected student feels safe biking *or* walking through Southside? Show your work; round to 2 decimals.

7.2 [1 point] What is the probability that all 8 students in the sample feel safe biking through Southside? Show your work; round to 2 decimals.

7.3 [1 point] What is the probability that at least one student does not feel safe biking through Southside? Show your work; round to 2 decimals.

7.4 [1 point] These calculations assume that each student's perspective on safety is *independent* of others. Explain why this assumption may not hold in real-world context. (1 sentence)

7.5 [2 points] City planners also found that 60% of students wear helmets while biking. Among helmet users, 95% feel safe, whereas only 50% of non-helmet users feel safe. If a randomly selected student reports feeling safe biking through Southside, what is the probability that they wear a helmet? Show your work; round to 2 decimals.

Q8 [6 Points]

In April 2009, the CDC detected a new influenza virus, later identified as H1N1, in two children in California.

As the outbreak spread, hospitals across the United States began screening patients with flu-like symptoms using a rapid influenza diagnostic test (RIDT). These tests were helpful for quick detection, but they were known to produce false results, especially when the disease was rare.

In one community hospital during the early stages of the outbreak:
- The prevalence of true H1N1 among patients with flu-like symptoms is 10%.
- The RIDT has 85% sensitivity, and 95% specificity.
- A total of 2,000 patients are tested.

	H1N1 Infected	Not H1N1 Infected	
Tests Positive	A	B	???
Tests Negative	C	D	???
Total	200	1800	2000

8.1 [2 points] Fill in the table with the expected number of patients in each cell (A-D).

- A:_____
- B:_____
- C:_____
- D:_____

8.2 [2 points] Using your completed table, calculate the Positive Predictive Value (PPV) and Negative Predictive Value (NPV) of the RIDT. Show your work, and round to two decimal places.

PPV :

NPV :

8.3 [2 points] Two months later, the prevalence of H1N1 among symptomatic patients increased to 40%. Explain how this change would affect the Sensitivity and Specificity of the test. (1-2 sentences).

Q9 [5 Points]

Lead exposure can harm children's health and development, even at low concentrations. Researchers from the National Health and Nutrition Examination Survey (NHANES) measured blood lead levels (BLLs) in U.S. children ages 1-5 years old. A random sample of 100 children had a mean BLL of 0.83 $\mu g/dL$, with a known population standard deviation of 0.40 $\mu g/dL$. Historically, the mean BLL among U.S. children was 1.0 $\mu g/dL$. Assume the blood lead levels are approximately Normally distributed.

9.1 [2 points] Calculate the standard error (SE) of the *sampling distribution* and construct a 95% confidence interval for the mean BLL. Show your work.

9.2 [2 points] Based on your confidence interval, would you reject or fail to reject the hypothesis that the population mean is 1.0 $\mu g/dL$? Why? (1 sentence)

9.3 [1 point] Researchers plan to repeat the study and want to increase statistical power to detect smaller differences in BLL. List one change in study design that would increase power.

10. [1 point] Extra Credit: If you lower the significance level from $\alpha = 0.05$ to $\alpha = 0.01$, what happens to the following?

Answer with: **increase**, **decrease**, or **no change**.

- Probability of Type II Error:_____
- Statistical Power:_____

Optional feedback

Take a moment to let us know any issues that came up on the exam. Or, draw us a picture!

This space is intended for scratch paper, we will NOT be grading this page